

# Eros Fani

📅 July 13, 1997

✉ eros.fani@gmail.com

🌐 erosfani.github.io

in LinkedIn

🔗 Github

## Professional and Teaching Experience

### Machine Learning Researcher, GENSYN

Oct 2025

→ Jul 2026

- Led research on **model merging**, **mixture of experts**, **LLMs routing**, and **decentralized ML**, resulting in **NeurIPS** and **ICLR workshop publications** [🔗](#) [🔗](#)
- Led development of **IR3DE-AXL**, an **open-source decentralized inference system** that routes requests to domain expert LLMs hosted by remote peers using Gensyn's AXL P2P network [🔗](#)
  - IR3DE-AXL allows decentralized peers to serve experts behind a unified chatbot Textual UI interface
  - Communication happens through AXL, a P2P decentralized networking layer that lets agents discover and communicate directly, exposing an HTTP API for fire-and-forget messaging and network topology discovery
  - Each peer locally constructs a lightweight IR3DE router to automatically select the most appropriate domain expert in the network for any given prompt. Peers can join or leave at any time while dynamically updating the router, and autonomously decide whether to exclude any expert from IR3DE routing by inspecting the statistics accessible through the UI
- Published technical articles on **decentralized LLMs inference routing** and **upcycling** of dense experts into **Mixture-of-Experts** architecture [🔗](#) [🔗](#)
- Invited as a speaker at two Gensyn X Space events [🔗](#) [🔗](#), each later featured in a Gensyn article [🔗](#) [🔗](#)

### Postdoctoral Fellow & Software Engineer, BASQUE CENTER FOR APPLIED MATHEMATICS (BCAM)

Nov 2024

→ Oct 2025

- Lead developer of the **Act.AI ERC proof-of-concept** (PI: Prof. N. Quadrianto)
  - Developed **Helios**, a framework for creditworthiness prediction on heterogeneous financial tabular data with LLMs and Extreme Gradient Boosting (XGBoost), in collaboration with the University of Sussex. Open data: ACS socioeconomic records by US states; closed, private data: from independent bank institutions
  - Implemented Parameter-Efficient Fine-Tuning (PEFT) Low-Rank Adaptation (LoRA) based techniques for post-training customer personalization

### Senior Researcher, PREDICTIVE ANALYTICS LAB, UNIVERSITY OF SUSSEX

May 2024

→ Oct 2025

- Led research on mitigating biases caused by **spurious correlations** in **federated learning**
- Developed **FedDiverse**, a new client selection method to improve performance by encouraging collaboration between diverse clients with minimal overhead, in collaboration with ELLIS Alicante

### MSc Thesis Supervisor, POLYTECHNIC UNIVERSITY OF TURIN

Sep 2022

→ Oct 2024

- Clustered Federated Learning project, resulted in a publication at **ICML 2025**
- Federated Visual Geo-Localization project, resulted in a publication at **FedVision @ CVPR 2024**

### Teaching Assistant, POLYTECHNIC UNIVERSITY OF TURIN

Sep 2022

→ Sep 2023

- Graduate courses: • Machine Learning and Deep Learning • Advanced Machine Learning • Data Analytics and Artificial Intelligence • Web Applications I
- Undergraduate course: • Computer Science

### Teaching Collaborator as an Undergrad, POLYTECHNIC UNIVERSITY OF TURIN

Sep 2018

→ Mar 2019

- Undergraduate courses: • Algorithms and Programming • Mathematical Analysis I • Mathematical Analysis II

## Education

### Ph.D. in Computer and Control Engineering, VISUAL AND MULTIMODAL APPLIED LEARNING LABORATORY (VANDAL), POLYTECHNIC UNIVERSITY OF TURIN

Nov 2021

→ Jul 2025

- Thesis title: **Addressing Heterogeneity in Federated Learning for Real-world Vision Applications**
- Publications at international peer-reviewed conferences and workshops, including **ICML**, **NeurIPS**, **CVPR**, **IROS**, and **WACV**, and to an **IEEE journal**
- Advisor: Prof. B. Caputo · Polytechnic University of Turin

### ELLIS Ph.D. Program - Academic Track, EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

Nov 2021

→ Jul 2025

- Primary Host: Prof. B. Caputo · Polytechnic University of Turin
- Exchange Host: Prof. N. Quadrianto · University of Sussex

### Visiting Period for the ELLIS Exchange, UNIVERSITY OF SUSSEX, EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

May 2024

→ Oct 2024

- Research Advisor: Prof. N. Quadrianto · University of Sussex

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| <b>Visiting Period</b> , ELLIS ALICANTE, EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)                                            | Sep 2024   |
| ◦ Research Advisor: Prof. N. Oliver · ELLIS Alicante                                                                                                 |            |
| <b>MSc in Data Science and Engineering</b> , POLYTECHNIC UNIVERSITY OF TURIN                                                                         | Oct 2019   |
| ◦ Final grade: <b>110/110 cum laude</b>                                                                                                              | → Oct 2021 |
| ◦ Thesis title: <b>On the Challenges of Class Imbalance in Federated Learning for Semantic Segmentation</b> , extended to an <b>IROS publication</b> |            |
| <b>BSc in Computer Engineering</b> , POLYTECHNIC UNIVERSITY OF TURIN                                                                                 | Oct 2016   |
| ◦ Final grade: <b>108/110</b>                                                                                                                        | → Jul 2019 |
| <b>Young Talent Project</b> , POLYTECHNIC UNIVERSITY OF TURIN, CRT FOUNDATION                                                                        | Oct 2016   |
| ◦ Study program of excellence open to only the <b>top 2% students</b> for the A.Y. 2016/17                                                           | → Jul 2019 |
| <b>High School Diploma</b> , LICEO SCIENTIFICO STATALE M. CURIE GIULIANOVA                                                                           | Sep 2011   |
| ◦ Final grade: <b>100/100 cum laude</b>                                                                                                              | → Jul 2016 |

## Publications

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| <b>E. Fani</b> , O. Ersoy. “ <i>IR3DE: A Linear Router for Large Language Models</i> ”. Resource-Adaptive Foundation Model Inference (AdaptFM) workshop of the International Conference on Machine Learning, 2026.                                     | AdaptFM<br>ICML26   |
| <b>E. Fani</b> , O. Ersoy. “ <i>Training-Free Dynamic Upcycling of Expert Language Models</i> ”. Scaling Post-training for LLMs (SPOT) workshop of the International Conference on Learning Representations, 2026.                                     | SPOT<br>ICLR26      |
| ◦ Extended version under review at a major ML conference.                                                                                                                                                                                              |                     |
| G. D. Németh, <b>E. Fani</b> , et. al. “ <i>FedDiverse: Tackling Data Heterogeneity in Federated Learning with Diversity-Driven Client Selection</i> ”. Third IEEE International Conference on Federated Learning Technologies and Applications, 2025. | FLTA25              |
| <b>E. Fani</b> , et. al. “ <i>Resource-Efficient Personalization in Federated Learning With Closed-Form Classifiers</i> ”. IEEE Access, 2025                                                                                                           | IEEE ACCESS         |
| A. Licciardi, ..., <b>E. Fani</b> , et. al. “ <i>Interaction-Aware Gaussian Weighting for Clustered Federated Learning</i> ”. Forty-second International Conference on Machine Learning, 2025.                                                         | ICML25              |
| <b>E. Fani</b> , et. al.“ <i>Accelerating Heterogeneous Federated Learning with Closed-form Classifiers</i> ”. Forty-first International Conference on Machine Learning, 2024.                                                                         | ICML24              |
| M. Dutto, ..., <b>E. Fani</b> , et. al. “ <i>Collaborative Visual Place Recognition through Federated Learning</i> ”. FedVision Workshop in Conjunction with the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024.                 | FEDVISION<br>CVPR24 |
| <b>E. Fani</b> , et. al. “ <i>Fed3R: Recursive Ridge Regression for Federated Learning with strong pre-trained models</i> ”. International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with NeurIPS, 2023.           | FL@FM<br>NEURIPS23  |
| <b>E. Fani</b> , et. al. “FedDrive v2: an Analysis of the Impact of Label Skewness in Federated Semantic Segmentation for Autonomous Driving”. 5th Italian Conference on Robotics and Intelligent Machines, 2023.                                      | I-RIM23             |
| D. Shenaj*, <b>E. Fani</b> *, et. al. “ <i>Learning Across Domains and Devices: Style-Driven Source-Free Domain Adaptation in Clustered Federated Learning</i> ”. IEEE/CVF winter conference on applications of computer vision, 2023.                 | WACV23              |
| L. Fantauzzo*, <b>E. Fani</b> *, et. al. “ <i>FedDrive: Generalizing Federated Learning to Semantic Segmentation in Autonomous Driving</i> ”. IEEE/RSJ International Conference on Intelligent Robots and Systems, 2022.                               | IROS22              |

\*Equal contribution.

## Selected Honors and Awards

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|---------------------------------------------------------------------------------------------|----------|
| <b>Invited to Present at the Flower AI Summit, Organized by Flower AI</b> <a href="#">🔗</a> | Apr 2026 |
| Presenting the Research Paper: TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS    |          |
| <b>Outstanding Reviewer award at ECCV24</b>                                                 | Sep 2024 |
| EUROPEAN CONFERENCE ON COMPUTER VISION 2024, EUROPEAN COMPUTER VISION ASSOCIATION (ECVA)    |          |

## Skills

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| ◦ <b>ML &amp; Research</b> : Efficient LLM inference, decentralized systems, peer-to-peer systems, data heterogeneity, tabular machine learning, semi-supervised learning, model evaluation and benchmarking, domain generalization and robustness, closed-form classifiers/ridge regression, Mixture-of-Experts, LLM inference routing, model merging, transformers, PEFT, LoRA, LLM post-training, federated learning, decentralized learning, distributed learning, semantic segmentation, image classification, domain adaptation, domain generalization and robustness, computer vision |
| ◦ <b>Programming &amp; Scripting</b> : Python, PyTorch, NumPy, Scikit-Learn, Weights & Biases, Matplotlib, Textual, Jax, FedJax, Haiku, Flax, Flower, C, HTML/CSS/Bootstrap, ReactJS, Vite, Java, C++, Bash, MATLAB, JavaScript, PHP, R                                                                                                                                                                                                                                                                                                                                                      |
| ◦ <b>Data &amp; Distributed Systems</b> : Apache Spark, Hadoop, SQL, AXL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| ◦ <b>Systems &amp; Infrastructure</b> : Linux, Git, HPC clusters, Slurm, Kubernetes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| ◦ <b>Languages</b> : Italian (native), English (professional), Spanish (elementary)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |